

TEMPORAL LOBE SEIZURES AND THE TECHNIC OF SUBTOTAL TEMPORAL LOBECTOMY*

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TEMPORAL LOBE SEIZURES are often amenable to treatment by radical operation. The abnormal, sclerotic area of cortex, which must be removed in most cases, lies in the deepest, the most inferior and mesial portion of the lobe where it is adjacent to the midbrain. This *incisural sclerosis* is evidently produced by herniation of this portion of the cortex through the incisura of the tentorium and compression of the arteries of supply at the time of birth. In spite of this fact, temporal lobe epilepsy most often presents itself in adults.

Five years ago Penfield and Steelman¹³ presented to this Association a discussion of the treatment of focal epilepsy in general by cortical excision. They pointed out that although the history of the entrance of surgery into this field had been filled with records of disappointment, the time had come when wise clinical analysis of the history of each case checked by electro-encephalographic localization and roentgen ray analysis made it possible for the surgeon to stand on firm ground in his approach to the radical treatment of selected patients.

Gibbs, Gibbs and Lennox⁷ reported that electro-encephalographic analysis of the "paroxysmal dysrhythmias" of epilepsy made it possible to recognize a very numerous group of cases to which they gave the name "psychomotor." The electro-

grams of these cases, they showed, were characterized by 4 to 6 per second waves, and initially they considered that the waves were generalized from all head regions.

Jasper⁹ and Jasper and Kershman¹⁰ pointed out that the discharge was localizable in the temporal regions, and that in the majority of cases an originating focal discharge was present in one temporal lobe.

Penfield and Flanigin¹¹ analyzed 68 cases of temporal lobe epilepsy in which excision of a larger or smaller portion of one temporal lobe had been carried out. The patients had been followed during a ten-year period ending in 1949. In 55 per cent of the cases the result was scored as success by patient and surgeon. About half of the patients in this success group had had no attacks following operation and half had had two or three attacks before apparent cessation of seizures.

Bailey and Gibbs,² also Green, Duisberg and McGrath⁸ have reported operations in temporal lobe epilepsy, removing anterior parts of one temporal lobe, the removal being carried out on the basis of electro-encephalography.

Our aim in the treatment of these patients by surgical excision is the same as it has been when the epileptogenic focus was situated elsewhere in the cerebral cortex, *i.e.*, to find and to remove the focus, as indicated by E.E.G., seizure pattern, electrical stimulation and the assistance of a conscious patient. We have learned to

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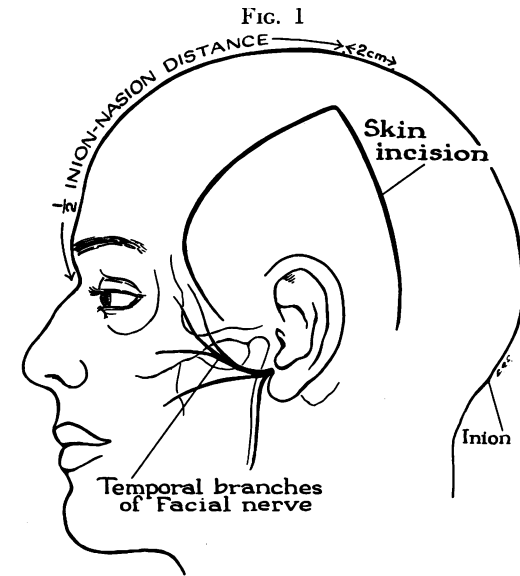


FIG. 2

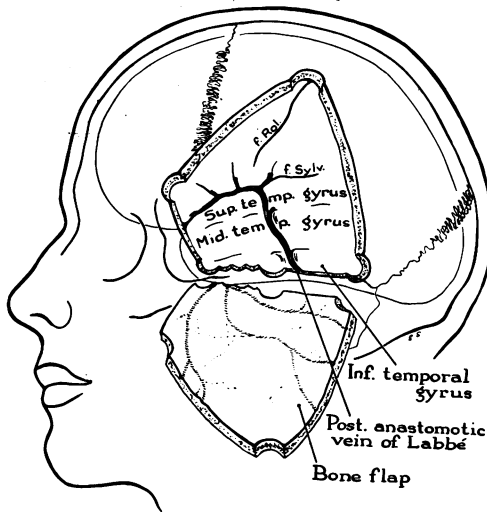
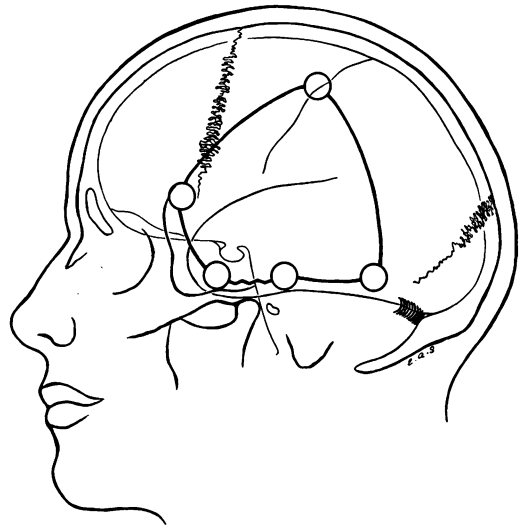


FIG. 3

FIG. 1.—Outline of the skin incision for subtotal temporal lobectomy.

FIG. 2.—The position of the burr holes for temporal craniotomy placed on a diagram of a lateral roentgenogram of the skull.

FIG. 3.—The bony opening designed to expose the fissures of Rolando and Sylvius as well as the frontal operculum.

expect a good therapeutic result only when these methods lead us to the removal of cortex which is abnormal by gross and microscopic standards. We will describe the surgical aspects of the operative procedure.

SUBTOTAL TEMPORAL LOBECTOMY*

A curved incision is made in the temporal region, as indicated in Figure 1. The scalp is then turned down separately. Burr holes are placed as low as the mastoid air cells will permit. It is well to use three holes so that the lower limb of the opening follows the floor of the middle fossa. If the mastoid air cells are opened, unfortunately, they may be closed at once with Horsley's bone wax. The other burr holes are so placed that a reasonable exposure of frontal and orbital operculum is made (Fig. 2), and the fissure of Rolando may be followed above the fissure of Sylvius for a little way into the angle of the bony opening (Fig. 3).

After the brain is exposed, stimulation is carried out. We use a bipolar or, preferably,

* The operation is best carried out under local anesthesia by the injection of 1-1500 Nupercaine in skin and temporal fossa and 1-4000 Nupercaine to the periosteum. Adrenalin is added to the analgesic solution. When dealing with young children or poorly adjusted adults, our anesthetist, Dr. André Pasquet, has produced sleep for us with intravenous Pentothal and nitrous oxide anesthesia. After the bone flap has been turned the patient is then allowed to wake up so that he can cooperate during electrical stimulation of the cortex and electrocorticography.

a unipolar electrode employing a square wave generator to supply the current. A strength of 1 to 2 volts with 2 milliseconds duration of wave and 60 pulses per second usually provides a threshold stimulus that will produce sensation when the electrode is applied to the postcentral gyrus and

near the area of abnormal epileptogenic cortex. In Case F. D. such a focus of E.C.G. abnormality was marked by the ticket A.

Excision of the temporal lobe is carried out along fissure lines so as to leave the pial covering and, therefore, normal circulation in the remaining banks of cortex. In the case

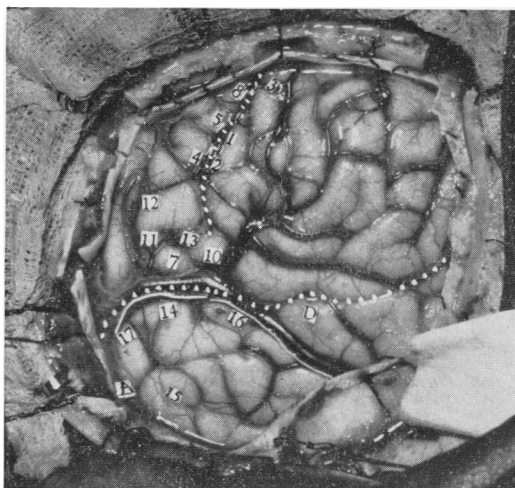


FIG. 4

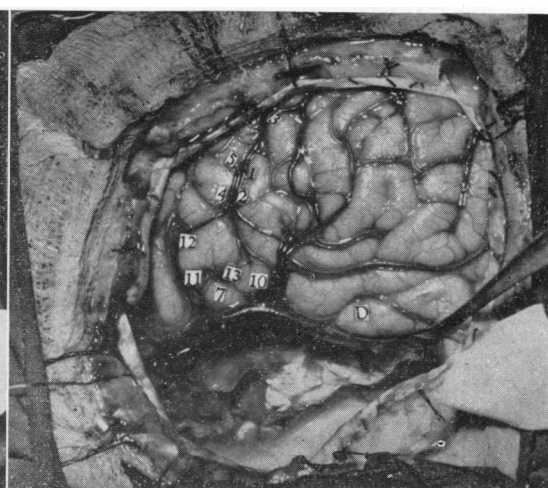


FIG. 5

FIG. 4.—Case F. D. Operative photograph showing stimulation tickets on the brain. The intended line of excision is shown by white thread parallel to the vein of Labbé. Dotted white lines are placed in the fissure of Rolando and in the fissure of Sylvius.

FIG. 5.—Case F. D. Partial excision of the temporal lobe has been carried out along the vein of Labbé. The white matter of temporal stem and the insula can be seen in the depth of the operative cavity.

movement on the precentral gyrus. Three or four volts may be used on the temporal lobe. When a positive response is obtained to stimulation, a small ticket is placed on the brain for convenient reference, as shown in Figure 4.

In Case F. D. (Fig. 4), stimulation at 1 and 2 produced sensation in the right side of the tongue, 3 in the right side of the face. Those numbered tickets therefore lie on the postcentral gyrus. Tickets 14, 15 and 16 lie on the temporal lobe. Stimulation there produced the confusion that was typical of the patient's habitual seizures.

Recording electrodes are placed on the brain and an electrocorticogram is carried out. Abnormal electrical discharge is thus usually recorded from electrodes placed

illustrated in Figure 4, the line of proposed excision, which follows the vein of Labbé, is shown by the white thread. The excised area is demonstrated in Figure 5, where the temporal stem of amputated white matter is seen distinctly. The insula lies above and anterior to the stem, with the middle cerebral arteries passing over it.

The line of lobe removal may, in the majority of cases, be carried along fissure lines, as mentioned above, and there are usually fissures crossing the first and second, and even the third temporal convolutions, obliquely. The anastomotic vein of Labbé, ordinarily runs in these crossing fissures, and it is usually better to follow this line in making the removal.

The excision is carried out by means of a

short sucker, the diameter of which should be 2 to 3 mm. With such a sucker it is possible to follow down the fissure of Sylvius to the insula and remove the temporal tissue which covers the insula without danger of injuring the middle cerebral artery complex.

must be opened during the excision. In such cases it has been our custom to close the ventricle by passing one or two silk sutures through the walls of the ependyma and tying them gently over a piece of Gel-foam. This closure of the ventricle is car-

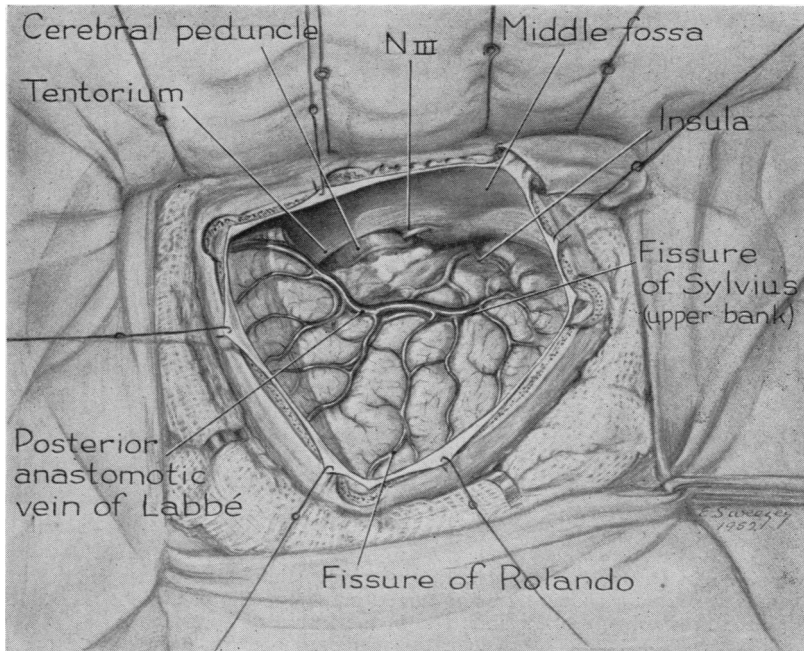


FIG. 6.—The brain is drawn upside down, as seen at operation, after temporal lobectomy. The relationship of the cerebral peduncle, third nerve and tentorial edge to the area of temporal excision is shown.

In the majority of cases the anterior portion of the first temporal convolution is found to be tough in the depth of the removal, and this tough tissue extends into the uncus and hippocampal gyrus mesially and posteriorly. The suction excision is carried along the lateral aspect of the midbrain (Fig. 6), great care being taken not to pass through the layer of pia which separates the operator from the third nerve and the posterior cerebral artery, as well as the lateral aspect of the midbrain. In this way the ablation may be carried back to include the full extent of the hippocampal gyrus.

When possible, it is best not to open the inferior horn of the ventricle, but often it

ried out so as to avoid postoperative aseptic meningitis (Finlayson and Penfield⁵). This form of reaction is due to the passage of elements of blood clot (probably hematin, as pointed out by Ira Jackson), into the ventricles and down into the basal cisterns.

When the anterior portion of the temporal lobe has been removed thus, it is well to carry out electrocorticography on the remaining cortex, as this may lead to the further discovery of an abnormal buried gyrus.

In certain cases it has seemed best to carry out removal of the insula (insulectomy). But the procedure is difficult and somewhat dangerous due to the fact that

the branches of the middle cerebral artery course over the insula. Thus, the portion that is exposed by removal of the temporal lobe can be removed without much difficulty, but the extension of this removal underneath the frontal operculum is more difficult because of danger of injury to the middle cerebral vessels.

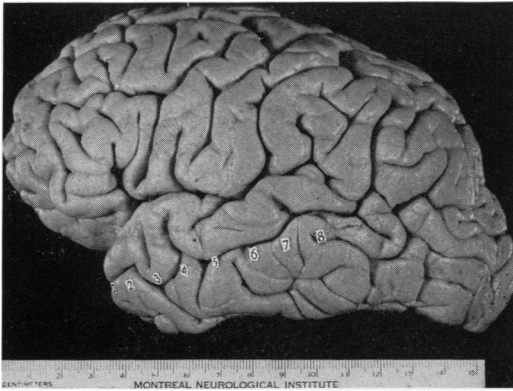


FIG. 7.—The mid-axis of the lateral temporal surface has been marked off in centimeters, beginning at the tip and passing posteriorly.

When the dura is first exposed, the middle meningeal artery is anesthetized with Nupercaine. After the dura has been opened, manipulation of the temporal lobe may be painful. This is apt to be due to elevation of the lobe, which exerts a pull upon the veins that cross to enter the venous sinuses below the lobe. Occasionally, also, electrical stimulation of the insula and in the vicinity of the middle cerebral arteries may cause pain because of spread of current to these vessels. The middle cerebral vascular tree is sensitive in about half of the patients studied, and is insensitive in the others.

To avoid all discomfort, the gasserian ganglion should be injected. This may be done during operation as follows: The anterior end of the lobe can usually be sucked out without pain. The floor of the middle fossa is then easily exposed and the ganglion identified by palpation with a

blunt instrument. A needle is inserted directly into the ganglion through the dura and Nupercaine is injected into it. This renders the patient's face numb, and the venous sinuses of the middle fossa and the middle cerebral arteries insensitive.

It has been our custom to close the dura completely at the end of the operation. A drain is usually placed in the extradural space and brought out through a stab wound in the scalp. This is ordinarily removed at the end of 24 hours. The bone flap is fastened in place with steel wire and the muscle, fascia, aponeurosis and scalp closed in layers.

OPERATIVE ANATOMY

The surgeon should be able to orient himself to deeper structures by examination of the lateral surface of the temporal lobe. He can do this by measuring from the tip of the lobe backward. In Figure 7, a brain is shown from which the pia has been removed and the distance from the tip backward along the mid-axis of the lateral surface has been marked off in centimeters.

In Figure 8, perpendicular transections of the temporal lobe of that brain have been made. The cross-sectional anatomy at 4, 5, 6 and 7 cm. is shown in the four lower squares of Figure 8 and in Figure 9. With this orientation it is possible to pass an electrode, coated except at the tip, into the temporal lobe and stimulate the uncus, amygdaloid nucleus or various portions of the superior temporal surface before deciding on excision. The electrode is introduced and then the current is switched on and off. An electrode, for example, passed directly inward in a plane 4 cm. from the temporal tip will enter the amygdaloid nucleus at a depth of 4 to 5 cm.

From this information it is obvious that operative excision of the temporal lobe anterior to the 4-cm. point on the lateral axis will not damage the ventricle. Excision posterior to this point will always cut across

the inferior horn. (Here it is assumed that all excisions go to the edge of the tentorium).

sucker after removal of the anterior end of the lobe. As already stated, only the mesial pia separates the dissecting sucker from

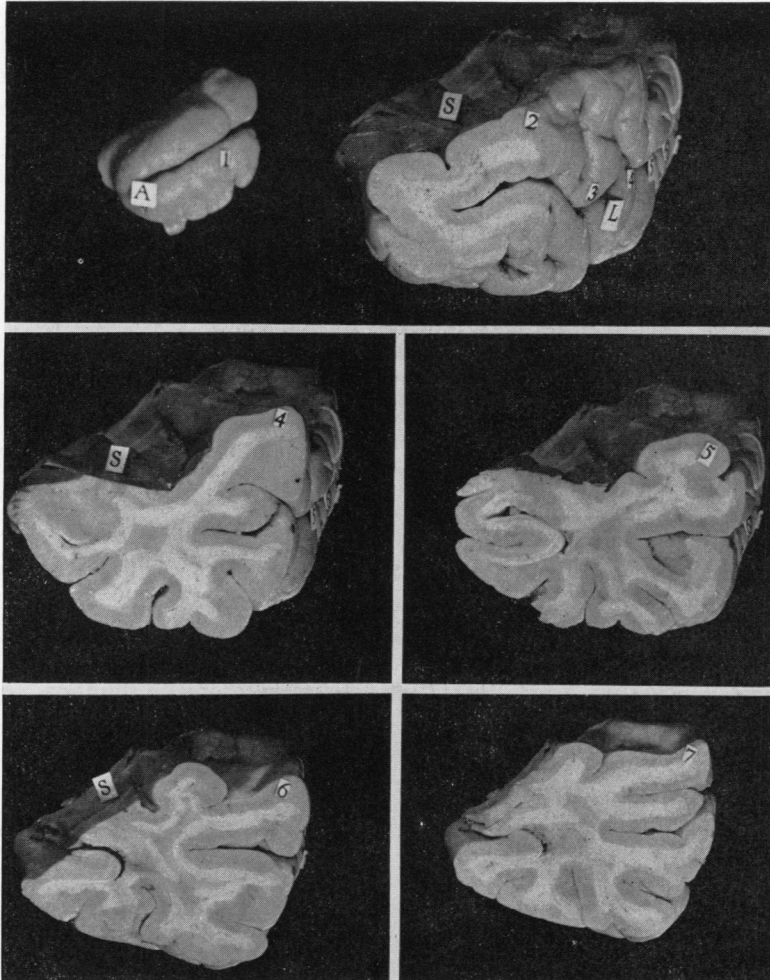


FIG. 8.—The cross-sections of the temporal lobe as marked off in centimeter segments along the mid-axis of the lateral surface in Figure 7. (A) anterior tip of lobe, (S) superior surface, (L) lateral surface.

The normal contour of the temporal horns is shown in Figure 10. This is a roentgenogram of a fixed brain with the base close to the tube. The ventricles have been injected with air and Pantopaque; the inferior horns are outlined in Pantopaque, the anterior horns in air. The brain stem is outlined by air in this illustration.

The hippocampal region which lies mesial to the inferior horn is removed by the

the peduncle, posterior cerebral artery, and the third nerve. These structures must be avoided by maintaining this pial boundary intact throughout the procedure.

Visual field defects are produced only by posterior temporal excisions. In our experience, excision anterior to or at the 5.5 cm. point, as measured in Figure 7, on the lateral axis does not result in permanent visual field defect. Excision posterior to

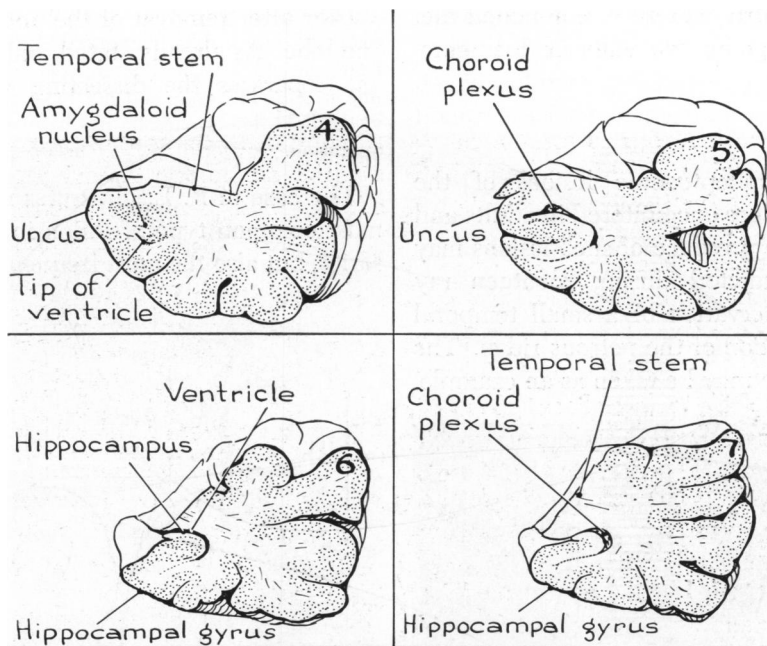


FIG. 9.—Drawings of the transections shown in Figure 8.



FIG. 10.—Roentgenogram of a fixed brain with the base close to the roentgen ray tube. The ventricles have been injected with air and Pantopaque. The inferior horns are outlined by Pantopaque, the rest by air.

this level results in permanent homonymous upper quadrantic defects. This conclusion is based on a study of the visual field defects which followed temporal lobec-

tomy in 30 cases. The portion of the visual radiation responsible for vision in the lower field passes around the temporal horn still farther posterior. The anatomical conclusions of this study will form the basis for a later report.

We have discussed the significant features of normal temporal lobe anatomy which influence our methods of exploration and excision. The extent of these excisions varies with the extent of the objective and electrographic abnormality and with the site of reproduction of initial features of the attack by stimulation.

We should turn now to the distribution and the significance of the objective abnormalities which are encountered.

ABNORMALITIES OF TEMPORAL CORTEX

Excision of normal cerebral cortex rarely, if ever, gives a good therapeutic result. The surgeon's aim must be to remove abnormal cortex which is also epileptogenic.

We have reviewed here the material excised in 157 cases of temporal lobe epilepsy. In 100 of these cases (63.6 per cent) the

lesion was atrophic and non-neoplastic. Some of the other patients had tumors, abscess scars, and vascular anomalies which are not in any way peculiar to the temporal lobe.

Atrophic, epileptogenic lesions of the temporal lobe are encountered in adults and children. The presence of such lesions may often be predicted during roentgen ray study, by observation of a small temporal fossa or elevation of the petrous ridge. The following case may be taken as an example.

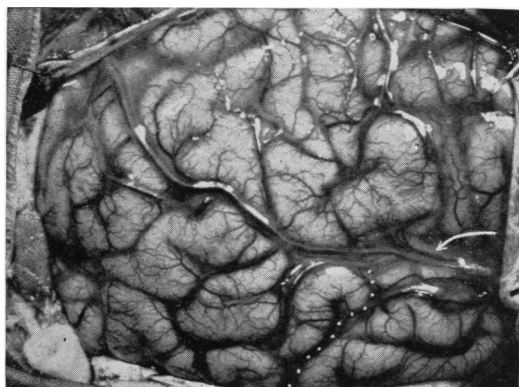


FIG. 11.—Case R. L., operative photograph. The arrow indicates an atrophic first temporal convolution. The white dots mark the line of excision.

Case 1.—R. L., a young housewife of 26 years suffered from cerebral seizures beginning in the right temporal lobe. The attacks were ushered in by initial perceptual illusions and had become quite frequent since their onset at the age of 25.

It seems likely that her birth was abnormal. She had had two generalized seizures at the age of 18 months, but after that no attacks until age 25. On roentgen ray examination it was found that the floor of her middle fossa was higher on the right than on the left. This may be taken as good evidence of a destructive lesion in the right hemisphere at birth or during infancy.

Right Temporal Craniotomy. At operation, the anterior part of the first temporal convolution appeared small and atrophic, as is indicated by the arrow in Figure 11. The whole right temporal lobe seemed small, but the most marked objective abnormality lay in the uncus, hippocampus and hippocampal gyrus. Here the tissue was yellowish, tough and avascular. The electrocorticographic localization coincided with the objective abnormality.

This lesion was removed in an excision of the temporal lobe which measured 4 cm. posteriorly from the tip on the lateral axis (see the dotted line in Fig. 11). She did not have a postoperative visual field defect and she has had no attacks since operation.

The case of R. L. illustrates not only the most common type of lesion in the present series, but also the most frequent location of

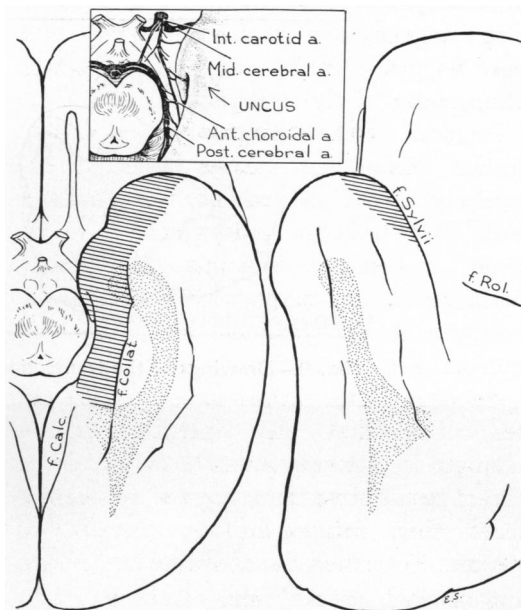


FIG. 12.—The most frequent location of objective abnormality (incisural sclerosis) is shown by the cross-hatching. The blood supply of those areas is indicated in the inset.

this lesion. This location is shown in Figure 12. In that figure the commonest site of incisural sclerosis is shown to be in the anterior portion of the first temporal convolution, the uncus, hippocampus and hippocampal gyrus.

PATHOLOGY

We can summarize the common gross characteristics of this lesion as toughness, yellowness and avascularity. At operation, these qualities are evident during dissection and removal. The abnormality may involve white as well as overlying gray matter.

The essential characteristics of the histology are: an astrogliosis, with piloid and giant astrocyte forms, decrease of neurones in the third cortical layer, an apparent decrease in number of cortical capillaries, and a thickening of the overlying pial vessels. This type of lesion may be of long duration, but the degenerating astrocytes, patchy demyelination, phagocytic cells and the microscopic cysts are evidence of a continuing degenerative process. Perhaps this may be produced by progressive vascular abnormality of the epileptogenic cortex.

Penfield and Humphreys¹² described similar lesions in various parts of the cerebral cortex of patients who suffered from focal epilepsy, lesions said to be the result of localized ischemia. They found evidence also of progressive scattered destruction in epileptogenic foci. This conclusion was reached after a study of 95 specimens from various areas of the cortex. Courville³ has described similar changes in different areas of cortex, as the direct result of ischemic anoxia. He emphasized their relationship to neonatal asphyxia. He called the abnormality "sclerosing cortical disease," and concludes that it is the minimal effect of anoxia (neonatal). Ford⁶ discussed such lesions with the conclusion that they are ischemic in origin.

We agree that these changes are ischemic in origin. But we also find evidence that when once established, there is further progressive destructive change over the course of many years. There is some experimental backing for this conclusion in the work of Windle, Becker and Weil.¹⁶ These workers studied the progressive changes which take place in guinea pig brains at successive intervals after natal ischemia. Their histologic findings are analogous to ours.

If this lesion is ischemic in origin, the pattern of its distribution in temporal cases should provide a clue to the mechanism of its production. In the majority of the cases

this cortical sclerosis has been found in the uncus and hippocampus and the anteromesial portion of the first temporal convolution, as indicated in Figure 12. In some cases it is found only in one of these regions; in others in all three. This portion of the first temporal convolution is supplied by middle cerebral branches which cross the tentorial edge. The uncus and hippocampus are supplied by the separate anterior choroidal artery, which must also cross the tentorium at its edge, and by branches from the posterior cerebral artery.

Spielmeyer¹⁵ and later Alexander and Putnam¹ have pointed out the singular susceptibility of the hippocampus and uncus to ischemic anoxia. They conclude that this susceptibility occurs because these structures possess only vessels of the fourth magnitude. This is particularly true in Ammon's horn.

We can present an additional reason for ischemia in these areas. All their vessels of supply cross the tentorial edge. Any pressure from above would compress them against this edge, producing temporary vascular occlusion and ischemia. If labor is prolonged or unduly rapid, the intracranial pressure of the fetus is increased and this produces uncal and hippocampal herniation. It results in "incisural sclerosis" (Earle, Baldwin and Penfield⁴). In the adult, high intracranial pressure frequently produces uncal herniation. This is a familiar phenomenon (Reid and Cone¹⁴). It may also occur in the infant at delivery. The herniating uncus compresses the anterior choroidal artery and some of the branches of the middle cerebral and posterior cerebral arteries against the tentorial edge. This would result in a temporary ischemia sufficient to produce the lesion we have described.

As already pointed out above, we found that in 100 out of 157 cases of temporal lobe excision, an atrophic lesion was present due to earlier localized ischemia. In nearly every

one of these cases the distribution of abnormality was that which would have been produced by temporal herniation through the incisura of the tentorium. The abnormality is therefore called incisural sclerosis and the technic of surgical excision must be planned to include the hippocampal region.

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DISCUSSION.—DR. WILLIAM JASON MIXTER, Boston: I don't want to let this paper go by without saying how much we owe Dr. Penfield for his work in this field. He has done a tremendous amount of work in it. He has shown us where the lesion lies, and I think most of us have had the feeling—I know I did—up to a short time ago that

the lesion lay much more on the surface of the temporal lobe rather than deeply, and that it could be picked up by electro-encephalography in the absence of any appearance of scar tissue. That is important, because partial operations in this region, with removal of the surface focus—I know from my own experience—do not give relief.